

CCA

CYBER SECURITY

SOCWatch

Wazuh SOC Monitoring Lab

Eight-Week Project-Based Internship Brief

Build a working SOC lab. Capture evidence. Explain your findings.

DURATION

8 weeks

PROJECT MODE

Individual

FINAL OUTPUT

SOC report + demo

CODEZELA CAREER ACCELERATOR

Your project at a glance

You will build one working security monitoring lab in eight weeks.

What you are building

A small Security Operations Centre lab using Wazuh on AWS. The lab will collect endpoint logs, show alerts and support basic security analysis.

Core result

- One Wazuh manager
- Three enrolled agents
- Active dashboard
- Captured alert evidence

How you will work

- Complete one weekly milestone
- Show real evidence in every sync
- Keep notes and screenshots organised
- Submit one final package

Build the core lab first. Advanced response features come only after the required alerts work.

The project is practical: setup, verify, test, document and explain.

The problem you will solve

Security teams need one place to see what is happening across different systems.

Current situation

- Logs are stored on separate machines
- Suspicious activity is hard to notice
- File changes may go unseen
- Vulnerable software may remain installed
- Evidence is difficult to prepare

Your SOC lab

- Collect logs from Linux and Windows
- View alerts in one dashboard
- Test file integrity monitoring
- Review vulnerability findings
- Prepare a clear technical report

The simple goal

Turn scattered endpoint activity into visible security evidence.

Do not try to build a huge enterprise SOC. Build a small lab that works clearly.

Target users and stakeholders

This project is a lab, but it should still be built for real operational roles.

1

SOC Analyst

- Reviews dashboard alerts
- Checks agent health
- Explains security findings

2

Cloud Engineer

- Creates AWS resources
- Controls network access
- Keeps infrastructure safe

3

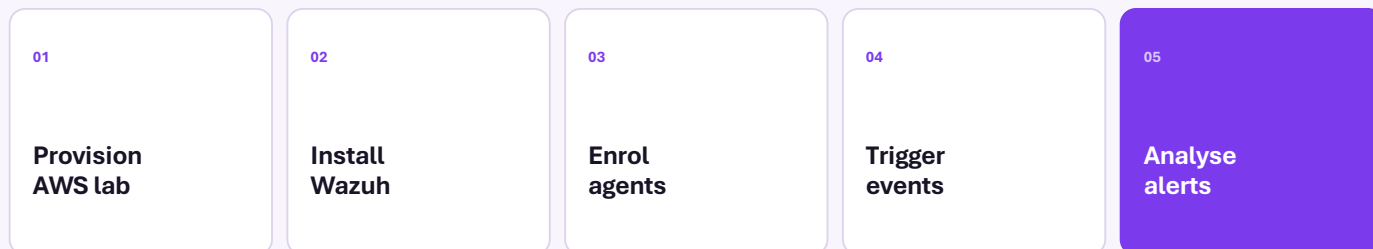
Mentor / Evaluator

- Reviews evidence
- Checks security rules
- Accepts final demonstration

Each student must have visible contribution evidence, even when the lab is built as a team.

From endpoint event to SOC evidence

This is the most important flow in the whole project.



Complete example

1. Team creates the AWS security group and four EC2 instances.
2. Wazuh manager, indexer and dashboard are installed on one server.
3. Linux and Windows agents are enrolled using the manager private IP.
4. Students trigger FIM, vulnerability and authentication test events.
5. Alerts are captured, explained and included in the final report.

Every mandatory task should create evidence that can be reviewed later.

Six areas must work

These areas form the minimum complete cyber security internship project.

01	AWS lab	Secure VPC, security group and four EC2 instances
02	Wazuh manager	All-in-one manager, indexer and dashboard running
03	Agent enrolment	Ubuntu, Rocky/RHEL and Windows agents active
04	Detection tests	Log ingestion, FIM, vulnerabilities and auth failures
05	Analysis evidence	Screenshots, findings and troubleshooting notes
06	Final report	Architecture, screenshots, findings and improvements

The project is complete only when the lab, alerts and report connect together.

AWS lab and security rules

The lab must be safe, controlled and easy for mentors to verify.

Item	Requirement
Instances	One manager and three endpoint agents
Network	Same VPC and subnet for internal routing
Admin access	SSH, HTTPS and RDP only from your public IP /32
Agent ports	1514 and 1515 allowed only from VPC CIDR
Evidence	Instance table with private IP, public IP and purpose

Security warning

Do not open 22, 443 or 3389 to 0.0.0.0/0.

Update your IP rule if your internet connection changes.

Required lab machines

Use small resources, but give the manager enough memory to run the dashboard.

Name	Role	OS	Minimum
wazuh-manager	Server, indexer, dashboard	Ubuntu 22.04 / 24.04	t3.medium, 30 GB
agent-linux-1	Endpoint 1	Ubuntu 22.04	t2/t3.micro, 8 GB
agent-linux-2	Endpoint 2	Rocky / RHEL 9	t2/t3.micro, 8 GB
agent-windows-1	Endpoint 3	Windows Server 2022	t3.small, 30 GB

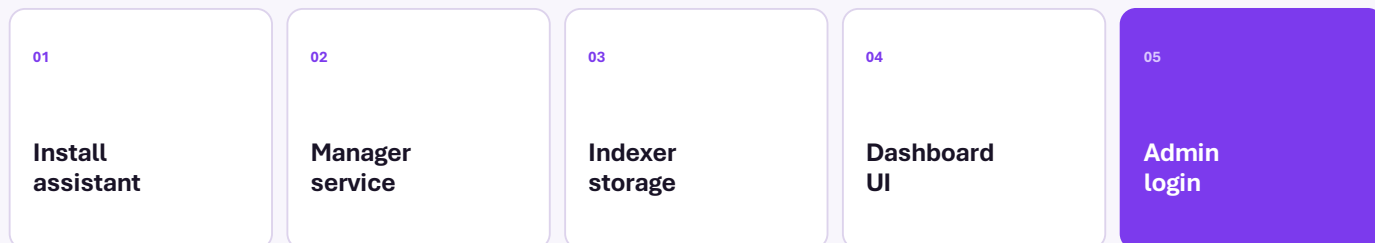
Critical detail

Agents must communicate with the manager private IP, not the public IP.

Take a screenshot or note the private IP immediately after creating the manager.

Central manager must run correctly

The central node is the heart of the SOC lab.



Required evidence

- Dashboard login screen
- Dashboard overview after login
- Running Wazuh services
- Recorded admin access method

Acceptance criteria

- Dashboard opens through HTTPS
- Admin can sign in
- Manager processes agent data
- No unnecessary public ports

Store credentials securely. Do not put passwords or keys in GitHub, LMS comments or screenshots.

Three agents must be active

The lab should show monitoring across different operating systems.

Agent	Operating system	Must show
agent-linux-1	Ubuntu	Active agent, logs and FIM test
agent-linux-2	Rocky / RHEL	Active agent and log events
agent-windows-1	Windows Server	Active agent and Windows event data

Ready when

The Wazuh dashboard shows all three agents with Active status.

Capture this screenshot before moving to detection testing.

Detection use cases to prove

Each test must produce a visible alert or dashboard finding.

01 Log ingestion Show Linux or Windows logs arriving in Security Events

02 File integrity monitoring Change a monitored file and capture the alert

03 Vulnerability detection Show CVE findings, severity or scan result

04 Authentication failure Generate failed SSH attempts and capture rules

05 Active response Stretch goal only after required evidence is complete

Do not claim a use case is complete without a screenshot and a short explanation.

Use one clear SOC lab setup

Keep the environment simple enough to complete and explain.

Area	Required choice
Cloud	AWS EC2 inside one VPC and subnet
SOC platform	Wazuh all-in-one manager, indexer and dashboard
Endpoints	Ubuntu, Rocky/RHEL and Windows Server agents
Access	SSH, RDP and HTTPS controlled by source IP
Evidence	Screenshots, diagrams, logs, report and presentation
Reference	Official current Wazuh documentation and approved brief

Use current official documentation. Avoid old third-party tutorials when versions differ.

Keep the architecture simple

One central Wazuh node collects security data from three endpoints.



Dashboard access uses HTTPS on 443. Agent data uses 1514. Enrolment uses 1515.

Port and access checklist

Every open port must have a reason and a limited source.

Port	Source	Purpose
22 SSH	Your public IP /32	Linux administration
443 HTTPS	Your public IP /32	Wazuh dashboard
3389 RDP	Your public IP /32	Windows setup
1514 TCP	VPC CIDR only	Agent security events
1515 TCP	VPC CIDR only	Agent enrolment

Reject during review

Any admin port opened to the whole internet without approval.

Evidence you must prepare

Screenshots should prove the lab is working, not decorate the report.

Required screenshots

- Dashboard login screen
- Dashboard overview
- Three active agents
- Log ingestion alert
- FIM alert
- Vulnerability result
- Authentication failure alert

Required notes

- Instance inventory
- Security group rules
- IP schema
- Troubleshooting log
- Short technical explanations
- Improvement plan
- Individual contribution summary

Blur or hide passwords, secret keys, tokens and full private credentials.

Make the lab safe and reviewable

These checks apply from Week 1, not only at the end.

Security baseline

- Least-privilege security groups
- No passwords in public files
- Key-based SSH for Linux
- Unused ports removed
- Instances stopped when not needed

Report baseline

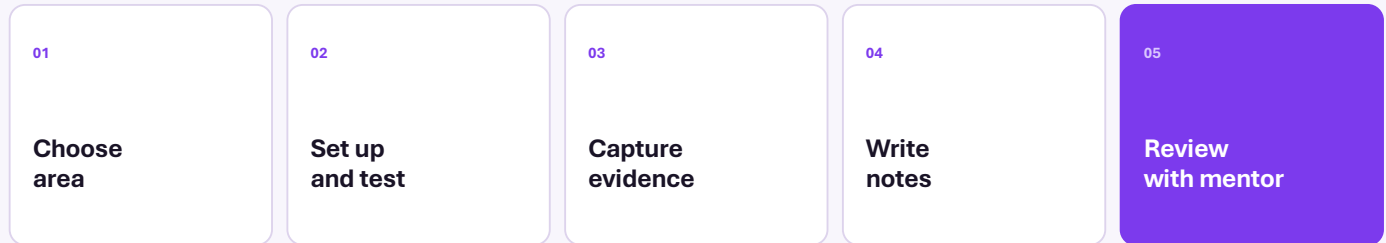
- Clear diagrams and IP table
- Screenshots labelled properly
- Each alert explained in plain words
- Two troubleshooting items recorded
- Individual contribution shown

Good evidence is simple

A reviewer should understand what happened without asking the team to explain every screenshot.

Build as a team, prove your own work

Every student should own a visible part of the lab and report.



Team evidence

- Shared evidence folder
- Shared report draft
- Weekly meeting updates
- Dashboard demonstration
- Final presentation

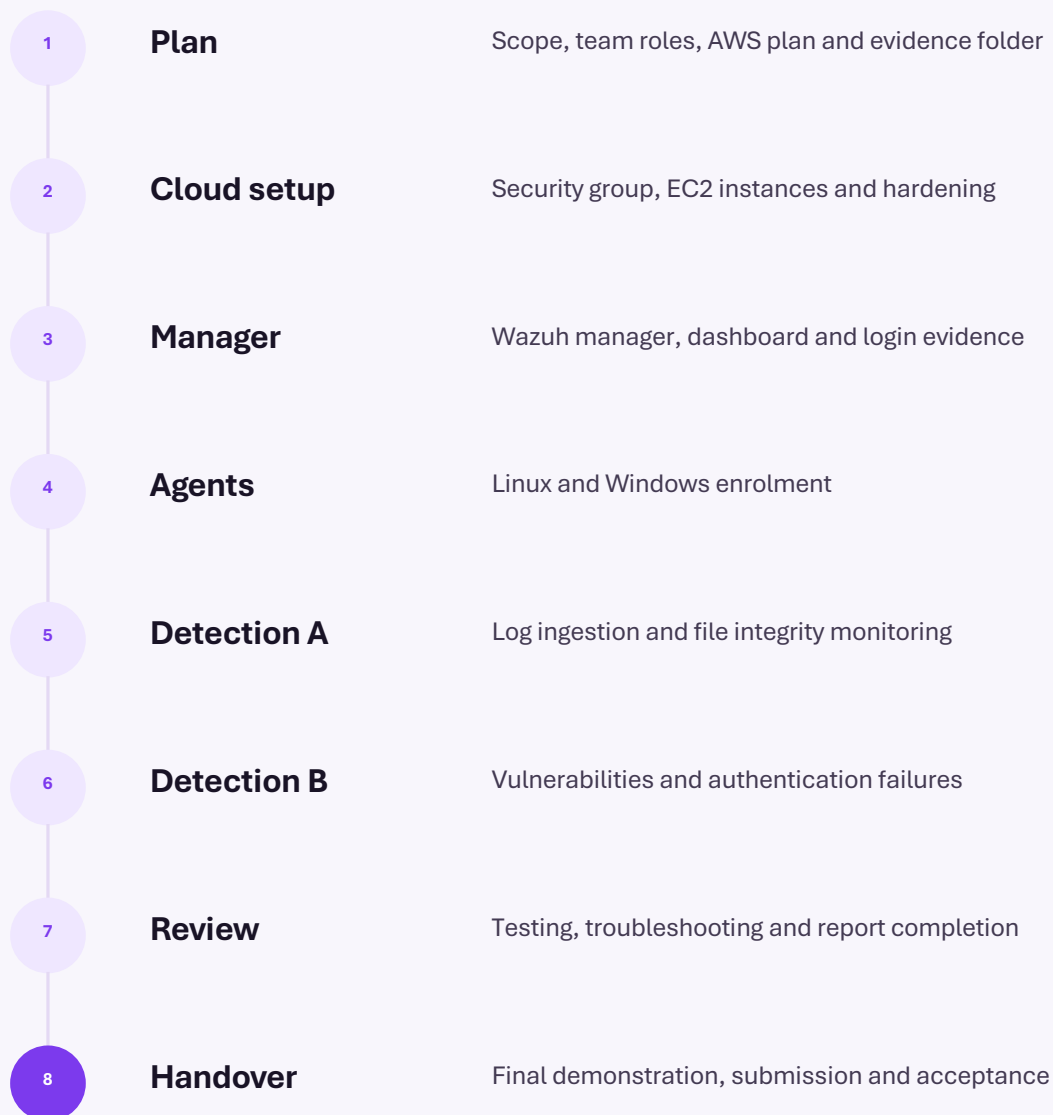
Individual evidence

- Own assigned tasks
- Screenshots captured by you
- Notes or section written by you
- Troubleshooting work explained
- Part of final defence

A single final upload does not show eight weeks of internship work.

Build the project week by week

Follow the sequence so later work is not blocked by unfinished foundations.



Understand, plan and set up

Finish the week with a clear lab plan, team ownership and evidence structure.

COMPLETE THIS WEEK

- Read the complete brief and list questions.
- Confirm team name, roles and ownership.
- Create the shared evidence folder and report outline.
- Draw the first AWS and Wazuh architecture diagram.
- Prepare the instance inventory template.
- Create the weekly update format.
- Confirm budget and free-tier safety rules.

Evidence to prepare

- Team roles
- Architecture draft
- Evidence folder
- Report outline
- Risk list

Show in the sync

- Explain the lab in one minute
- Show the architecture draft
- Confirm each member ownership

WEEKLY CHECKPOINT

Mentor approves the scope, team ownership, evidence structure and AWS setup plan.

Create the AWS foundation

Finish the week with secure cloud infrastructure ready for Wazuh deployment.

COMPLETE THIS WEEK

- Create the security group with approved inbound rules.
- Launch the manager and three endpoint instances.
- Record private IP, public IP, OS and instance type.
- Confirm all instances are in the same VPC and subnet.
- Harden Linux SSH access where required.
- Test SSH, RDP and HTTPS access rules.
- Start the troubleshooting log.

Evidence to prepare

- Security group screenshot
- Instance inventory
- Connection proof
- Hardening notes
- Troubleshooting log

Show in the sync

- Show the SG rules
- Show instance list
- Explain why ports are limited

WEEKLY CHECKPOINT

All required machines exist, access is controlled and no admin port is open to the whole internet.

Deploy the Wazuh manager

Finish the week with Wazuh manager, indexer and dashboard running on the central server.

COMPLETE THIS WEEK

- Connect to the manager server safely.
- Install Wazuh using the approved all-in-one method.
- Retrieve and store admin credentials securely.
- Open the dashboard through HTTPS.
- Capture login and overview screenshots.
- Record Wazuh version and installation notes.
- Add manager setup steps to the report.

Evidence to prepare

- Dashboard login
- Dashboard overview
- Wazuh version
- Setup notes
- Service status

Show in the sync

- Log in to Wazuh
- Show overview page
- Explain manager, indexer and dashboard

WEEKLY CHECKPOINT

Dashboard is accessible, admin login works and the central Wazuh services are running.

Enrol Linux and Windows agents

Finish the week with all three endpoints listed as Active in Wazuh.

COMPLETE THIS WEEK

- Install the Ubuntu agent.
- Install the Rocky or RHEL agent.
- Install the Windows Server agent.
- Use the manager private IP for enrolment.
- Start and enable the agent services.
- Verify agents from CLI or dashboard.
- Capture the active agents screenshot.

Evidence to prepare

- Three active agents
- Agent service proof
- OS notes
- Private IP mapping
- Issue log

Show in the sync

- Show the Agents page
- Filter by each agent
- Explain manager private IP usage

WEEKLY CHECKPOINT

All three endpoint agents are visible in Wazuh with Active status.

Prove logs and file integrity monitoring

Finish the week with basic telemetry and one clear FIM alert captured.

COMPLETE THIS WEEK

- Open Security Events and filter by agent name.
- Verify Linux or Windows logs are arriving.
- Configure FIM monitoring on a safe test directory.
- Create or modify a test file.
- Find the FIM alert in the dashboard.
- Capture screenshots with labels.
- Write a short explanation of hashing and file changes.

Evidence to prepare

- Log event screenshot
- FIM alert screenshot
- Config note
- Technical explanation
- Updated report

Show in the sync

- Show live events
- Show FIM alert
- Explain what changed and how Wazuh detected it

WEEKLY CHECKPOINT

At least one raw log event and one FIM event are captured and explained.

Prove vulnerability and authentication alerts

Finish the week with two more security use cases captured and explained.

COMPLETE THIS WEEK

- Confirm vulnerability detection is enabled.
- Allow scans to run or use an approved vulnerable package test.
- Capture CVE or scan-result evidence.
- Generate several failed SSH login attempts safely.
- Find authentication failure or brute-force rules.
- Record source IP and rule details.
- Update troubleshooting and findings.

Evidence to prepare

- Vulnerability evidence
- Auth failure evidence
- Rule IDs or details
- Risk notes
- Findings draft

Show in the sync

- Show CVE or scan result
- Show failed-login alert
- Explain the security meaning

WEEKLY CHECKPOINT

Vulnerability and authentication-failure evidence are visible and included in the report.

Finalise, test and document

Finish the week with the full report ready for mentor review.

COMPLETE THIS WEEK

- Check every required screenshot is available.
- Complete the architecture and IP schema.
- Write technical summaries in simple language.
- Complete at least two troubleshooting entries.
- Add production improvement ideas.
- Review security group rules again.
- Prepare the final presentation.

Evidence to prepare

- Full report draft
- Diagram and IP table
- Troubleshooting log
- Improvement plan
- Presentation draft

Show in the sync

- Walk through the report
- Show missing-item checklist
- Run through the presentation plan

WEEKLY CHECKPOINT

Mentor confirms the project is ready for final demonstration after any requested fixes.

Present, demonstrate and hand over

Finish the internship by demonstrating the lab and submitting the final package.

COMPLETE THIS WEEK

- Open with the problem and lab architecture.
- Show the Wazuh dashboard and agent health.
- Demonstrate or replay each detection use case.
- Explain findings, risks and troubleshooting.
- Each student explains their own contribution.
- Submit report, links and screenshots through LMS.
- Clean up or stop cloud resources after approval.

Evidence to prepare

- Final report
- Final slides
- Dashboard demo
- Evidence folder
- Submission confirmation

Show in the sync

- Present the lab flow
- Answer technical questions
- Show individual contribution

WEEKLY CHECKPOINT

Final project is accepted when the lab evidence, report and individual defence are satisfactory.

Use the same update format every week

Students should prepare the update before the mentor meeting.

1 **Work completed**

2 **Demonstrable evidence**

3 **Tasks in progress**

4 **Blockers or risks**

5 **Decisions required**

6 **Individual contributions**

7 **Plan for next week**

A sync is not only a verbal update. Students must show dashboards, screenshots, diagrams or report progress.

Do not skip approval points

Major stages need mentor review before the team moves too far ahead.

Checkpoint	Due	Acceptance
Project plan approval	End of Week 1	Scope, roles and diagram approved
AWS foundation approval	End of Week 2	Security rules and instances approved
Manager approval	End of Week 3	Dashboard login and overview approved
Agent approval	End of Week 4	Three active agents approved
Detection review	Weeks 5 to 6	Required alerts and explanations approved
Final readiness review	End of Week 7	Report and presentation ready
Final acceptance	Week 8	Demo, defence and submission accepted

What the team must submit

Deliverables are grouped so students know what belongs where.

Group deliverables

- AWS instance inventory
- Security group screenshots
- Architecture diagram
- Wazuh dashboard evidence
- Detection screenshots
- Final report
- Final presentation

Individual deliverables

- Assigned task summary
- Screenshots or notes contributed
- Troubleshooting item explained
- Section of report written
- Final defence contribution

Portfolio-ready outputs

A clean SOC lab report, architecture diagram and use-case evidence that can be discussed in interviews.

Submission package

WhatsApp is not the official submission method. Use the LMS or approved portal.

Item	Must include
Final report PDF	Architecture, IP schema, screenshots, summaries, troubleshooting
Presentation PDF/PPTX	Short final demonstration deck
Evidence links	Cloud folder or repository link if used
Access details	Dashboard URL and demo credentials if still available
Screenshots	Clearly labelled and placed inside the report
Individual summary	Each member contribution and defence topic

Do not submit secret keys, private SSH keys or real passwords.

How the project will be assessed

Students must be able to explain their own work and findings.

Area	Weight
Working SOC lab and completeness	30%
Cyber security execution and alert analysis	20%
Weekly milestones and working process	15%
Individual contribution	15%
Documentation and evidence	10%
Final demonstration and defence	10%

AI may assist learning and writing, but unexplained copied work will not prove internship participation.

Completion checklist

Use this page before submitting the final package.

- ☐ AWS security rules are safe and documented
- ☐ Wazuh dashboard works or evidence is captured
- ☐ Three agents are Active in screenshots
- ☐ Log ingestion, FIM, vulnerability and auth-failure evidence are included
- ☐ Troubleshooting log has at least two resolved issues
- ☐ Architecture diagram and IP table are clear
- ☐ Each student contribution is stated
- ☐ Final report and presentation are submitted through LMS